

CM Descent for One-Place Archimedean Stark Invariants: Major-Revision Working Draft

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Abstract

This is a major-revision working draft, not a submission manuscript and not a source of proved packet claims. It isolates the precise cyclic-quartic normalization used in a proposed descent from one-place real-quadratic invariants to Stark's proved theorem over imaginary quadratic fields. The remaining proof obligations are an explicit Fourier-to-CM-norm bridge and, for the $\mathbb{Q}(\sqrt{6})$ example, a prime-ideal and finite-valuation analysis of the auxiliary Euler factors. Previously computed packets are retained as candidates until those obligations close.

1 Status and separation of automorphisms

Let k be imaginary quadratic and let E/k be cyclic quartic, with

$$G = \text{Gal}(E/k) = \langle \sigma \rangle \simeq C_4, \quad \tau = \sigma^2.$$

The involution τ fixes k . It is therefore *not* global complex conjugation, which acts nontrivially on k . Whenever E is stable under global complex conjugation, denote the latter by j in the normal closure and put $E^+ = E^{(j)}$. The internal anti-unit relation concerns τ ; the positive CM norm concerns j . They must not be conflated.

Fix a faithful ray character with $\psi(\sigma) = i$, and let S contain the complex place and all conductor primes. Suppose Stark's proved imaginary-quadratic rank-one theorem applies and $|S| \geq 3$, so that it supplies a global unit $\varepsilon \in \mathcal{O}_E^\times$, unique modulo $\mu(E)$. Put $e = |\mu(E)|$ and

$$\ell_g = \log |g\varepsilon|_{\text{ord}}.$$

Lemma 1 (Cyclic-quartic normalization). *Assume in addition that*

$$\tau(\varepsilon) \equiv \varepsilon^{-1} \pmod{\mu(E)}. \quad (1)$$

Then

$$\zeta'_S(0, g) = -\frac{2}{e} \ell_g \quad (2)$$

and, with $L_S(s, \psi) = \sum_{g \in G} \overline{\psi(g)} \zeta_S(s, g)$,

$$L'_S(0, \psi) = -\frac{4}{e} (\ell_1 - i\ell_\sigma). \quad (3)$$

These coefficients are independent of the numerical value of e .

Proof. Stark's normalized complex absolute value is $|z|_w = |z|_{\text{ord}}^2$. His formula $\zeta'_S(0, g) = -e^{-1} \log |g\varepsilon|_w$ gives (2). Condition (1) gives $\ell_{\sigma^2} = -\ell_1$ and $\ell_{\sigma^3} = -\ell_\sigma$. Since

$$(\overline{\psi(1)}, \overline{\psi(\sigma)}, \overline{\psi(\sigma^2)}, \overline{\psi(\sigma^3)}) = (1, -i, -1, i),$$

Fourier inversion of (2) gives (3). □

Lemma 2 (Torsion invariance). *If j acts as complex conjugation on roots of unity, then the ordered magnitudes $(|g\varepsilon|)_{g \in G}$ and positive products*

$$q_g = (g\varepsilon) j(g\varepsilon) \quad (4)$$

are unchanged when ε is multiplied by an element of $\mu(E)$.

Proof. Magnitudes are unchanged by roots of unity. If $\xi \in \mu(E)$, then $j(\xi) = \xi^{-1}$, so the additional factor in (4) is $\xi j(\xi) = 1$. $\square$

2 The missing Fourier-to-norm bridge

Equality of induced representations proves equality of the relevant Artin L -functions, but does not by itself identify an individual one-place invariant. The packet results require the following proposition, which is a proof obligation rather than a theorem of this draft.

Proposition 3 (Required bridge; not yet proved). *Let θ be a selected one-place ray character over the real quadratic field K , and suppose*

$$\mathrm{Ind}_{G_K}^{G_{\mathbb{Q}}} \theta \simeq \mathrm{Ind}_{G_k}^{G_{\mathbb{Q}}} \psi \quad (5)$$

as linear representations, with all imprimitive local Euler factors specified. For a ray class A , an explicitly determined group-ring element $b_A \in \mathbb{Z}[G]$ and sign $\epsilon_A \in \{\pm 1\}$ should satisfy

$$\log X_A = \epsilon_A \log \left| N_{E/E^+} \left(\varepsilon^{b_A} \right) \right|. \quad (6)$$

The map $A \mapsto b_A$, all local factors, and the Artin labels must be derived by Fourier inversion rather than inferred from matching polynomials.

To prove Proposition 3, the revision must print:

1. both Fourier inversions, including the one-place sign class;
2. the exact character correspondence arising from (5);
3. every local Euler factor at primes removed from either side;
4. the resulting integral group-ring coefficient b_A ;
5. the exact normal-closure map taking (4) to the asserted one-place ray field; and
6. the Artin action on every displayed packet value.

3 The auxiliary-prime gate for $\mathbb{Q}(\sqrt{6})$

The current computation establishes two rational-prime Euler polynomials, normalized analytic runs at $q = 3, 5$, exact unit-lattice divisibility, and the cross-route identity $q_8^3 = q_{12}^2$. Those facts do not yet eliminate a residual S -unit ambiguity.

Before the case can be promoted, the revision must, for each imaginary quadratic base and each rational q :

1. factor $q\mathcal{O}_k$ and name every prime $\mathfrak{q}$ added to S ;
2. state whether $\mathfrak{q}$ is split, inert, or ramified and whether it divides the conductor;
3. compute $\psi(\mathfrak{q})$ and the group-ring Euler factor $1 - \mathrm{Frob}_{\mathfrak{q}}^{-1} N\mathfrak{q}^{-s}$;

4. compute the finite valuations of the enlarged Stark unit at all primes above q ; and
5. invoke a precise uniqueness theorem showing that equality of the archimedean logarithms and those finite valuations leaves only a root of unity.

Only after these checks may a statement such as “the enlarged unit is the square of the natural unit modulo torsion” enter a proof.

4 Candidate corpus, without promotion

The existing computations for the $\mathbb{Q}(\sqrt{35})$ packet, the first $e = 6$ tranche, RQ-000458, and $\mathbb{Q}(\sqrt{6})$ remain useful candidate data and regression controls. They are not claimed as theorems in this draft. A future submission will require Proposition 3 and, for $\mathbb{Q}(\sqrt{6})$, the complete auxiliary-prime gate above.